



7. Diabetes Technology: Standards of Care in Diabetes—2026

American Diabetes Association
Professional Practice Committee for
Diabetes*

Diabetes Care 2026;49(Suppl. 1):S150–S165 | <https://doi.org/10.2337/dc26-S007>

The American Diabetes Association (ADA) “Standards of Care in Diabetes” includes the ADA’s current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA’s clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

Diabetes technology is the term used to describe the hardware and software that people with diabetes use to assist with self-management, ranging from lifestyle modifications to glucose monitoring and automated therapy adjustments. Historically, diabetes technology has been divided into two main categories: insulin administration (insulin given by syringe, pen, inhalation, patch devices, or pump [also called continuous subcutaneous insulin infusion]) and glucose measurement (assessment by blood glucose monitoring [BGM] or continuous glucose monitoring [CGM]). Diabetes technology now also includes automated insulin delivery (AID) systems that use CGM-informed algorithms to modulate insulin delivery. Furthermore, diabetes technology encompasses connected insulin pens and diabetes self-management support software that serve as medical devices. Diabetes technology, coupled with education, follow-up, pharmacotherapy as needed, and support, can improve the lives and health of people with diabetes; however, the complexity and rapid evolution of the diabetes technology landscape can also be a barrier to implementation for people with diabetes, their care partners, and the health care team.

GENERAL DEVICE PRINCIPLES

Recommendations

7.1 Diabetes devices should be offered to people with diabetes. **A**

7.2 The type(s) and selection of devices should be individualized based on a person’s specific needs, circumstances, preferences, and skill level. In the setting of an individual whose diabetes is partially or wholly managed by someone else (e.g., a young child or a person with cognitive impairment or dexterity, psychosocial issues, and/or physical limitations), the caregiver’s skills and preferences are integral to the decision-making process. **E**

7.3a When prescribing a continuous glucose monitoring (CGM) device, ensure that people with diabetes and caregivers are offered initial and ongoing training and education as indicated by individual circumstances. Education should

*A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at <https://doi.org/10.2337/dc26-SINT>.

Duality of interest information for each contributor is available at <https://doi.org/10.2337/dc26-SDIS>.

Suggested citation: American Diabetes Association Professional Practice Committee for Diabetes. 7. Diabetes technology: Standards of Care in Diabetes—2026. *Diabetes Care* 2026;49(Suppl. 1): S150–S165

© 2025 by the American Diabetes Association. Readers may use this article as long as the work is properly cited, the use is educational and not for profit, and the work is not altered. The version of record may be linked at <https://diabetesjournals.org/care>, but ADA permission is required to post this work on any third-party site or platform. This publication and its contents may not be reproduced, distributed, or used for text or data mining, machine learning, or similar technologies without prior written permission. Requests to reuse, adapt, or distribute this work may be sent to permissions@diabetes.org. More information is available at <https://diabetesjournals.org/journals/pages/license>.

include utilization of data, including uploading or sharing data to monitor and adjust therapy. **C**

7.3b When prescribing an automated insulin delivery (AID) system, people with diabetes and their caregivers must receive education on how to use and troubleshoot the system. This education should occur at regular intervals as needed. Education should include utilization of the integrated system and its data, including uploading or sharing data to monitor and adjust therapy. **C**

7.4 Health care professionals working with people with diabetes should be aware of available technologies and seek additional support when needed. **E**

7.5 People with diabetes using CGM, continuous subcutaneous insulin infusion (CSII), and/or AID for diabetes management should have continued access to devices across third-party payors, regardless of age or A1C levels. **E**

7.6 Children and adolescents should be supported at school in the use of diabetes technology, such as CGM systems, CSII, connected insulin pens, and AID systems. **E**

7.7 For adults with diabetes using diabetes technology, reasonable accommodations in educational and work settings should include having sufficient time to manage their devices and respond to high and low glucose levels. **E**

7.8 Consider early initiation, including at diagnosis, of CGM, CSII, and AID depending on a person's or caregiver's needs and preferences. **C**

7.8a There should be no requirement of C-peptide level, **B** the presence of islet autoantibodies, **B** or duration of insulin treatment **C** before initiation of CSII or AID.

7.9 Standardized reports for all CGM, CSII, AID, and connected insulin devices with a minimum of a single-page report, such as the standardized CGM report and weekly summary, should be available and utilized. Options for daily and weekly reports and raw data should be available. **E**

Technology is rapidly changing, and there is no one-size-fits-all approach to technology use in people with diabetes.

Insurance coverage can lag behind device availability, people's interest in devices and willingness for adoption can vary, and health care teams may have challenges in keeping up with newly released technology. An American Diabetes Association resource, which can be accessed at diabetes.org/living-with-diabetes/treatment-care/diabetes-technology-guide, can help health care professionals and people with diabetes make decisions on the initial choice of device(s). Other sources, including device manufacturers, can help people troubleshoot when difficulties arise (1–7).

Education and Training

In general, no device used in diabetes management works optimally without education, training, and ongoing support. There are multiple resources, including online tutorials and training videos as well as written material, on the use of devices. People with diabetes and their caregivers vary in comfort level with technology, and some prefer in-person training and support. Those with more education regarding device use have better outcomes (1,2); therefore, the need for additional education should be periodically assessed, particularly if treatment goals are not being met.

Training on the use of CGM often differs from that of pump and AID systems. In some settings, CGM training can be done by users themselves with company-provided training materials (for instance, people are able to use the over-the-counter [OTC] CGM brands on their own), although for underresourced, younger (8,9), and older (10) individuals, more repetition and time spent reviewing concepts is often indicated. Additionally, regular monitoring and review of the data obtained from CGM devices is needed to inform and optimize clinical care (8). Education and training should be available to any person who needs it but should not constitute a barrier to device use in a motivated individual.

Insulin pumps and AID systems, on the other hand, generally require training and education for safe use. There are studies where youth with type 1 diabetes have been able to self-initiate tubeless AID systems (11), but for most, training with a certified or trained diabetes educator and education specialist is necessary for full understanding and safe use of the technology.

Health care professionals should be aware of the availability of diabetes devices and their recommended uses and how to access the resources needed to use them. Given the rapidly changing landscape of technology, it is expected that software and/or hardware are being constantly updated; hence, periodic education is helpful to keep health care professionals informed and updated. For those interested in more advanced involvement with device use, there are defined competencies described as basic, fundamental, intermediate, and advanced that are specific to the role of each health care team member (12). The health care team's knowledge and competency are even more critical when people with diabetes are started on advanced diabetes technologies, such as AID systems. In such situations, training of the team is vital and should include a discussion about realistic expectations for the ability of the initiated system to achieve glucose goals, fundamental differences from the previous treatment tools, the system's features and limitations, and the best way to use the new system to maximize the benefits it can offer (13).

Use in Schools for Children and Adolescents

Technology use should be supported in school settings, and instructions for device use should be outlined in the student's diabetes medical management plan (DMMP). A backup treatment plan should be included in the DMMP for potential device failure. School nurses and designees should complete training to stay up to date on diabetes technologies prescribed for use in the school setting. Updated resources to support diabetes care at school, including training materials and a DMMP template, can be found online at (<https://diabetes.org/advocacy/safe-at-school-state-laws/diabetes-medical-management-plan>).

Use for Adults

For adults using devices in educational and employment settings, reasonable accommodations should be provided to allow time and space necessary for safe device use (14). This includes time to respond appropriately to high and low glucose levels and deal with any device notifications and issues.

Initiation of Device Use

Diabetes technology should be initiated soon after the time of diagnosis and anytime thereafter, as indicated by the user's needs and wishes. The use of CGM and BGM devices should be started from the outset of the diagnosis of diabetes, particularly in people who require insulin for management (3,4,15,16). CGM use allows for close tracking of glucose levels with adjustment of insulin dosing and lifestyle modifications and removes the burden of frequent BGM. In addition, early CGM initiation after diagnosis of type 1 diabetes in children and adolescents has been shown to decrease A1C levels and is associated with high parental satisfaction and reliance on this technology for diabetes management (5,7). Training on alarm and alert settings when initiating CGM is crucial to avoid alarm fatigue or overload. Interruption of access to CGM is associated with a worsening of outcomes (6,17); therefore, it is important for individuals on CGM to have consistent access to devices. A recent systematic review with meta-analysis also supports the use of CGM in people with type 2 diabetes not using insulin, with observations of improved glycemic outcomes and individual experience, reduced health care resource utilization, and acceptable cost-effectiveness (18).

AID systems are the preferred insulin delivery system in individuals with type 1 diabetes and type 2 diabetes on multiple daily injections (MDI) and people with other forms of insulin-deficient diabetes. They can be considered for use in people on basal insulin who are not reaching their targets. Early initiation of AID therapy has been shown to be beneficial. In an open-label, multicenter, randomized, parallel clinical trial enrolling youth with newly diagnosed type 1 diabetes, initiation of an AID system within 21 days from diagnosis showed 10% higher time in range (TIR) (70–180 mg/dL [3.9–10.0 mmol/L]) and lower A1C at 12 months versus usual care (15). In addition, use of diabetes technology overall improves A1C and increases the number of people achieving an A1C <7% (19). There are no data that the presence or absence of C-peptide or islet autoantibodies is associated with responsiveness to continuous subcutaneous insulin infusion or AID therapy. People with type 2 diabetes who continue to produce insulin

have been shown to benefit from AID therapy (20–23). Similarly, the duration of diabetes is not related to outcomes with the use of technology. Therefore, none of these measures should be used when determining suitability for continuous subcutaneous insulin infusion or AID therapy.

Standardized Reporting

Use of a standardized CGM tracing is helpful for people with diabetes and clinicians (24) and in multiple settings, from connected insulin pens (25) to CGM (26) to AID systems (27). Ideally, both people with diabetes and their health care teams can access and analyze the data, both between and at clinic visits to inform self-management and medication dose titration.

BLOOD GLUCOSE MONITORING

Recommendations

7.10 People with diabetes should be provided with blood glucose monitoring (BGM) devices as indicated by their circumstances, preferences, and treatment. People using CGM devices must also have access to BGM at all times. **A**

7.11 People who are taking insulin and using BGM should be encouraged to check their blood glucose levels when appropriate based on their insulin therapy. This may include checking when fasting, prior to meals and snacks, after meals, at bedtime, in the middle of the night, prior to, during, and after exercise, when hypoglycemia is suspected, after treating low blood glucose levels until achieving normoglycemia, when hyperglycemia is suspected, and prior to and while performing critical tasks such as driving. **B**

7.12 Health care professionals should be aware of the differences in accuracy among blood glucose meters. Only meters approved by the U.S. Food and Drug Administration (FDA) (or comparable regulatory agencies for other geographical locations) with proven accuracy should be used, with unexpired test strips purchased from a pharmacy or licensed distributor and properly stored. **E**

7.13 Although BGM in people on noninsulin therapies has not consistently shown clinically significant reductions in A1C levels, it may be

helpful when modifying meal plans, physical activity plans, and/or medications (particularly medications that can cause hypoglycemia) in conjunction with a treatment adjustment program. **E**

7.14 Consider potential interference of medications and substances on glucose levels measured by blood glucose meters. **B**

Major clinical trials of insulin-treated people with diabetes have included BGM as part of multifactorial interventions to demonstrate the benefit of intensive glycemic management on diabetes complications (28). BGM is thus an integral component of effective therapy for individuals using insulin. In recent years, CGM has emerged as a method for the assessment of glucose levels that improves outcomes compared with BGM in many settings (discussed below). Glucose monitoring (BGM or CGM) allows people with diabetes to evaluate their individual responses to therapy and assess whether glycemic goals are being safely achieved. Integrating glucose testing results into diabetes management can be a useful tool for guiding medical nutrition therapy and physical activity, preventing hypoglycemia, or adjusting medications (particularly prandial insulin doses or correction bolus doses). The specific needs and goals of the person with diabetes should dictate BGM and CGM frequency and timing. As recommended by the device manufacturers and the U.S. Food and Drug Administration (FDA), people with diabetes using CGM must also have access to BGM for multiple reasons, including whenever there is suspicion that the CGM is inaccurate, while waiting for the CGM device to warm up, when there is a disruption in CGM transmission, for calibration (if needed) or if a warning message appears, when CGM supplies are delayed, and in any clinical setting where glucose levels are changing rapidly (>2 mg/dL/min), which could cause a discrepancy between CGM and blood glucose values.

Meter Standards

Glucose meters meeting FDA guidance for meter accuracy provide the most reliable data for diabetes management. There are several current standards for the accuracy of blood glucose meters,

but the two most used are those of the International Organization for Standardization (ISO) (ISO 15197:2013) and the FDA (**Table 7.1**). In the U.S., currently marketed meters must meet the standard under which they were approved, which may not be the current standard. Moreover, the monitoring of current accuracy postmarketing is left to the manufacturer and not routinely checked by an independent source.

People with diabetes assume their glucose meter is accurate because it is FDA cleared, but that may not be the case. In prior analyses with glucose meters, 14 of 18 glucose meters met the minimum accuracy requirements (29). However, few studies have compared meters head-to-head.

Certain meter system characteristics, such as the use of lancing devices that are less painful (30) and the ability to reapply blood to a strip with an insufficient initial sample, or meters with integrated speech that can read aloud glucose levels for visually impaired individuals (31), may also be beneficial to people with diabetes (32) and may make BGM less burdensome to perform.

Use of Strips From Unlicensed Entities

People with diabetes should be advised against purchasing or reselling preowned or secondhand test strips, as these may give incorrect results. Only unopened and unexpired vials of glucose test strips should be used to ensure BGM accuracy.

Optimizing Blood Glucose Meter Use

Optimal use of BGM devices requires proper review and interpretation of data by both the person with diabetes and the health care professional to ensure

that data are used in an effective and timely manner. In a large cohort analysis of over 24,000 adults with both type 1 and type 2 diabetes, more frequent use of BGM was associated with lower A1C levels, with greater BGM frequency associated with lower A1C in people with type 1 diabetes across the lifespan (33,34). In young adults with type 1 diabetes, there was also a correlation between greater BGM frequency and lower A1C levels (34). Among those who check their blood glucose at least once daily, many reported taking no action when results were high or low (35). Some meters now provide advice to the user in real time when monitoring glucose levels (36), whereas others can be used as a part of integrated health platforms (37). People with diabetes should be taught how to use BGM data to adjust food intake, physical activity, or pharmacologic therapy to achieve their treatment goals. The ongoing need for and frequency of BGM should be reevaluated at each routine visit to ensure its effective use (34,38).

People With Diabetes Treated With Intensive Insulin Therapies

Glucose monitoring is particularly important for people with diabetes treated with insulin therapy to detect and prevent hypoglycemia and hyperglycemia. Most individuals on intensive insulin therapies (MDI or insulin pump therapy) should be encouraged to assess glucose levels using BGM (and preferably CGM with BGM backup) prior to meals and snacks, at bedtime, occasionally postprandially, prior to, during, and after physical activity, when they suspect hypoglycemia or hyperglycemia, after treating hypoglycemia until they are

normoglycemic, and prior to and while performing critical tasks such as driving. For many individuals using BGM alone, this requires doing finger sticks 6–10 times daily, although individual needs may vary. A database study of almost 27,000 children and adolescents with type 1 diabetes showed that, after adjusting for multiple confounders, increased daily frequency of BGM was significantly associated with lower A1C levels (–0.2% per additional check per day) and with fewer acute complications (39).

People With Diabetes Using Basal Insulin and/or Oral Agents and Noninsulin Injectables

The evidence is insufficient regarding when to prescribe BGM and how often monitoring is needed for insulin-treated people with diabetes who do not use intensive insulin therapy, such as those with type 2 diabetes taking basal insulin with or without oral agents and/or noninsulin injectables. However, for those taking basal insulin, assessing fasting glucose with BGM to inform dose adjustments to achieve blood glucose goals results in lower A1C levels (40).

In people with type 2 diabetes not taking insulin, studies have not consistently shown benefit, although the most benefit has been seen in individuals participating in structured clinical programs (41,42). Some individuals find BGM useful to provide insight into the impact of nutrition, physical activity, and medication management on glucose levels. Glucose monitoring may also be useful in assessing hypoglycemia, glucose levels during intercurrent illness, or discrepancies between measured A1C and glucose levels when there is concern an A1C

Table 7.1—Comparison of ISO 15197:2013 and FDA blood glucose meter accuracy standards

Setting	FDA*	ISO 15197:2013*
Hospital use	95% within 12% for BG ≥ 75 mg/dL 95% within 12 mg/dL for BG < 75 mg/dL 98% within 15% for BG ≥ 75 mg/dL 98% within 15 mg/dL for BG < 75 mg/dL	95% within 15% for BG ≥ 100 mg/dL 95% within 15 mg/dL for BG < 100 mg/dL 99% in A or B region of consensus error grid†
Home use	95% within 15% for all BG in the usable BG range‡ 99% within 20% for all BG in the usable BG range‡	

BG, blood glucose; FDA, U.S. Food and Drug Administration; ISO, International Organization for Standardization. To convert mg/dL to mmol/L, see endmemo.com/medical/unitconvert/Glucose.php. *Data shown in the FDA column are from the FDA (191). Data shown in the ISO column are from the FDA (192). †The range of blood glucose values for which the meter has been proven accurate and will provide readings (other than low, high, or error). ‡Values outside of the “clinically acceptable” A and B regions are considered “outlier” readings and may be dangerous to use for therapeutic decisions (193).

result may not be reliable in specific individuals (for more details, see section 2, “Diagnosis and Classification of Diabetes”). A key consideration is that performing BGM alone does not lower blood glucose levels. To be useful, the information must be integrated into clinical and self-management treatment plans. Further, as noted above, a recent systematic review with meta-analysis did support use of CGM in people with type 2 diabetes with benefits to glycemic management, individual experience, health care resource utilization, and cost-effectiveness (18,43).

Glucose Meter Inaccuracy

Although many meters function well under various circumstances, health care professionals and people with diabetes must be aware of factors that impair meter (and CGM) accuracy. A meter reading that seems discordant with the clinical picture needs to be retested or tested in a laboratory. Health care professionals in intensive care unit settings need to be particularly aware of the potential for incorrect meter readings during critical illness, and laboratory-based values should be used if there is any doubt. Some meters give error messages if meter readings are likely to be false (44).

Oxygen. Most currently available glucose monitors use an enzymatic reaction linked to an electrochemical reaction, either glucose oxidase or glucose dehydrogenase (45). Glucose oxidase monitors are sensitive to the oxygen available and should only be used with capillary blood in people with normal oxygen saturation. Higher oxygen tensions (i.e., arterial blood or oxygen therapy) may result in false low-glucose readings, and low oxygen tensions (i.e., high altitude, hypoxia, or venous blood readings) may lead to falsely elevated glucose readings. Glucose dehydrogenase-based monitors are generally not sensitive to oxygen.

Temperature. Because the reaction is sensitive to temperature, all monitors have an acceptable temperature range (45). Most will show an error if the temperature is unacceptable, but a few will provide a reading and a message indicating that the value may be incorrect.

Interfering Substances. There are several physiologic and pharmacologic factors that interfere with glucose readings measured

with either personal blood glucose meters or professional blood glucose meters used in various inpatient settings (neonatal intensive care units, hospital wards, and intensive care units) (45). Meters vary in terms of their sensitivity to interfering substances, and this information should be available in the prescribing information for the meter. Some of the possible interfering substances are listed in **Table 7.2**.

CONTINUOUS GLUCOSE MONITORING DEVICES

See **Table 7.3** for definitions of types of currently available CGM devices.

Recommendations

7.15 Use of CGM is recommended at diabetes onset and anytime thereafter for children, adolescents, and adults with diabetes who are on insulin therapy, **A** on noninsulin therapies that can cause hypoglycemia, **C** and on any diabetes treatment where CGM helps in management. **C** The specific CGM device and method for use should be made based on the individual's circumstances, preferences, and needs. **E**

7.16 In people with diabetes on insulin therapy, CGM devices should be used as close to daily as possible for maximal benefit. **A** People with diabetes should have uninterrupted access to their supplies to minimize gaps in CGM. **A**

7.17 During pregnancy for individuals with type 1 diabetes, CGM can help achieve glycemic goals (e.g., time in range and time above range) **A** and A1C goal **B** and may be beneficial for other types of diabetes in pregnancy. **E** See section 15, “Management of Diabetes in Pregnancy,” for more detail regarding use of technology in pregnancy.

7.18 In circumstances when consistent use of CGM is not feasible, consider periodic use of personal or professional CGM to adjust medication and/or lifestyle. **C**

7.19 Skin reactions, either due to irritation or allergy, should be assessed and addressed to aid in successful use of devices. **E**

7.20 People who wear CGM devices should be educated on potential interfering substances and other factors that may affect accuracy. **C**

CGM measures interstitial glucose (which correlates well with plasma glucose, although at times, it can lag if glucose levels are rising or falling rapidly). Although historically there were two basic types of personal CGM devices, real-time CGM (rtCGM) and intermittently scanned CGM (isCGM), now most available prescribed CGM devices have similar characteristics: no swiping required,

Table 7.2—Some of the more common interfering substances and/or conditions that affect blood glucose meters (for inpatient and outpatient use)

Substance or condition	Potential effects on glucose readings measured by BGMs*
Maltose†	Falsely higher blood glucose readings
Galactose	Falsely higher blood glucose readings
Xylose	Falsely higher blood glucose readings
N-Acetylcysteine	Falsely higher or lower blood glucose readings (depending on BGM design)
Acetaminophen	Falsely higher or lower blood glucose readings (depending on BGM design)
Dopamine	Falsely higher or lower blood glucose readings (depending on BGM design)
Pralidoxime (2-PAM)	Falsely higher or lower blood glucose readings (depending on BGM design)
Hydroxyurea	Falsely higher or lower blood glucose readings (depending on BGM design)
Vitamin C	Falsely higher or lower blood glucose readings (depending on BGM design)
Hematocrit (high)	Falsely lower blood glucose readings
Hematocrit (low)	Falsely higher blood glucose readings

*These are potential effects. There are blood glucose monitors (BGMs) that behave differently than listed in this table. Refer to product labeling for product-specific information. †Unmodified glucose dehydrogenase pyrroloquinoline quinone (GDH/PQQ) enzyme method only. Modern BGM designs do not incorporate unmodified GDH-PQQ enzyme.

Table 7.3—Continuous glucose monitoring devices

Type of device	Brand*	Availability	Alarms
rtCGM	Libre 2 Plus and Libre 3 Plus	Prescription	Yes
	Dexcom G6 and G7	Prescription	Yes
	Eversense 365	Prescription	Yes
	Guardian 4	Prescription	Yes
	Simplera	Prescription	Yes
OTC-CGM	Dexcom Stelo	OTC	No
	Abbott Lingo	OTC	No
Professional CGM	Abbott FreeStyle Libre Pro	In office	No
	Dexcom G6 Pro	In office	No

CGM, continuous glucose monitoring; isCGM, intermittently scanned CGM; OTC, over the counter; rtCGM, real-time CGM. *Generic names not available.

continuous stream of data, and adjustable alarms and alerts. If an individual is still using a CGM that requires scanning, the limitations of those systems should be addressed. There are two categories of CGM devices generally available for personal use: those prescribed for diabetes management (rtCGM) and over-the-counter CGM (OTC-CGM), which lacks alarms and alerts and can be used by people not taking diabetes medications that increase the risk of hypoglycemia. Anyone can purchase the OTC-CGM devices, including those without diabetes or with prediabetes who wish to assess their glycemic responses to their lifestyles, including the effects of food choices and exercise. Finally, there are what has been called professional CGM devices, which are devices owned and applied by health care professionals that generate glucose data (blinded or not) over a short period of time.

Table 7.3 provides definitions for the types of CGM devices. For people with type 1 diabetes using CGM, frequency of sensor use is an important predictor of A1C lowering for all age-groups (46,47).

Few real-time systems require calibration by the user, and the need to recalibrate varies in frequency depending on the device. However, all CGM systems require use of BGM to confirm accuracy, especially in settings where there is perceived discordance between sensor reading and the individual's perceived sense of their clinical situation.

Most CGM systems are designated as integrated CGM (iCGM) and cleared as a class II medical device by the FDA for integration with other digitally connected devices. The exact CGM that integrates with any given device (e.g., automated insulin system or connected insulin pen)

varies, and available options should be explored with individuals when choosing an integrated system.

Benefits of Continuous Glucose Monitoring

Data From Randomized Controlled Trials

Multiple randomized controlled trials (RCTs) have been performed using CGM devices, and the results have largely been positive in terms of reducing A1C levels and/or episodes of hypoglycemia if participants regularly wore the devices (46–53). The benefits of CGM have been shown regardless of age, sex, education or income levels, or baseline diabetes characteristics.

The initial studies were done primarily in adults, children, and adolescents with type 1 diabetes on insulin pump therapy and/or MDI (46,47,50,51,54). The primary outcome was met and showed benefit in adults of all ages (46,55,56), including older individuals (57–59). Data in children show that CGM use in young children with type 1 diabetes reduced hypoglycemia; in addition, behavioral support of parents of young children with diabetes using CGM showed the benefits of reducing hypoglycemia concerns and diabetes distress (46,51,60). Similarly, A1C level reduction was seen in adolescents and young adults with type 1 diabetes using rtCGM (50).

RCT data on CGM use in individuals with type 2 diabetes on MDI (61) or insulin (including basal insulin) plus noninsulin therapies (61–64) show consistent reductions in A1C levels and increases in TIR (70–180 mg/dL [3.9–10 mmol/L]) and time below range but not a reduction in rates of level 3 hypoglycemia (65). RCT data for CGM benefits in people with type 2 diabetes not using insulin are increasing and generally have

shown greater benefits of CGM compared with BGM for A1C, TIR, time below range, and time above range (TAR) as well as greater user-reported satisfaction (66,67). In addition, CGM benefits were reported in a population of adults with type 2 diabetes (on noninsulin and insulin-including therapy) with reduction in A1C levels, increase in TIR, and reduction of time in hyperglycemia (>180 mg/dL [>10 mmol/L] and >250 mg/dL [>13.9 mmol/L]) (62). Although glycemic metrics are improved, patient-reported outcomes have shown less consistent satisfaction in this individual population.

Although short-term use of CGM in youth with type 2 diabetes did not impact short-term glucose changes or A1C improvement, users reported behavioral changes with increased blood glucose measurements, increased insulin administration, and overall improved diabetes management and quality of life (68,69). The improvements in type 2 diabetes have largely occurred without changes in insulin doses or other diabetes medications. CGM discontinuation in individuals with type 2 diabetes on basal insulin caused partial reversal of A1C reduction and TIR improvements, suggesting that continued CGM use achieves the greatest benefits (17).

Observational and Real-world Studies

CGM systems are widely available in many countries for people with diabetes, and this allows for the collection of large amounts of data across groups of people with diabetes.

Data for isCGM in adults with diabetes include results from observational studies, retrospective studies, and analyses of registry and population data (70,71). Glycemic benefits in people with type 1 and type 2 diabetes (on MDI, basal insulin, and noninsulin therapies) have been seen in prospective and retrospective studies (49,72).

Reductions in acute diabetes complications, such as diabetic ketoacidosis (DKA), episodes of severe hypoglycemia or diabetes-related coma, and hospitalizations for hypoglycemia and hyperglycemia, have been observed in adults with type 1 or type 2 diabetes (73–75), with persistent effects observed even after 2 years of CGM initiation (76). Similar reductions of acute diabetes events and all-cause inpatient hospitalizations were seen in a retrospective review of

adults with type 2 diabetes treated with basal insulin or with noninsulin therapy 6 months after initiation of isCGM (77).

Retrospective data from rtCGM use in adults (78) with type 1 or type 2 diabetes treated with insulin showed that the use of rtCGM significantly lowered A1C levels and reduced rates of emergency department visits or hospitalizations for hypoglycemia but did not significantly lower overall rates of emergency department visits, hospitalizations, or hyperglycemia.

Recent data have emerged from a real-world observational analysis of rtCGM use in adults with type 2 diabetes not treated with insulin. In this study, rtCGM benefits were observed at 6 months and 12 months versus baseline, with reduction of mean glucose levels, reduction of glucose management indicator (GMI), increase in TIR, increase in time in tight target range (70–140 mg/dL [3.9–7.8 mmol/L]), and reduction in TAR >180 mg/dL (>10 mmol/L) and >250 mg/dL (>13.9 mmol/L) (79).

Real-time Continuous Glucose Monitoring Device Use in Pregnancy

CGM indication has expanded to include pregnancy for Dexcom G7, FreeStyle Libre 2, and FreeStyle Libre 3, which enhances care in this population (80,81). A discussion of CGM use in pregnancy can be found in section 15, “Management of Diabetes in Pregnancy.”

Use of Professional Continuous Glucose Monitoring and Intermittent Use of Personal Continuous Glucose Monitoring Professional CGM devices, which provide retrospective data, either blinded or unblinded, for analysis can be used to identify patterns of hypoglycemia and hyperglycemia (82–84). Professional CGM

can be helpful to evaluate an individual’s glucose levels when either rtCGM or isCGM is not available to the individual or they prefer a blinded analysis or a shorter experience with unblinded data. It can be particularly useful in individuals using agents that can cause hypoglycemia, as the data can be used to evaluate periods of hypoglycemia and make medication dose adjustments if needed. It can also be useful to evaluate periods of hyperglycemia.

Some data have shown the benefit of intermittent use of CGM (rtCGM or isCGM) in individuals with type 2 diabetes on noninsulin and/or basal insulin therapies (63,85). In these RCTs, people with type 2 diabetes not on intensive insulin therapy used CGM intermittently compared with those randomized to BGM. Both early (63) and late (63,86) improvements in A1C levels were found.

Furthermore, in a real-world study, the use of professional CGM in individuals with type 2 diabetes not on insulin at baseline and at 6 months of follow-up resulted in lower A1C at 6 months as well as a shift toward greater use of glucose-lowering medications with cardiometabolic benefits, such as sodium–glucose transporter 2 inhibitors and glucagon-like peptide 1 receptor agonists (87). Use of professional or intermittent CGM should always be coupled with analysis and interpretation for people with diabetes along with education, as needed, to adjust medication and change lifestyle behaviors (88–90).

Side Effects of Continuous Glucose Monitoring Devices

Contact dermatitis (both irritant and allergic) has been reported with all devices that attach to the skin (91–93). In some cases, this has been linked to

the presence of isobornyl acrylate, a skin sensitizer that can cause an additional spreading allergic reaction (94). It is important to ask CGM users periodically about adhesive reactions, as tape formulations may change over time. Patch testing can sometimes identify the cause of contact dermatitis (95). Identifying and eliminating tape allergens is important to ensure the comfortable use of devices and promote self-care (96–99). The PANTHER Program offers resources in English and Spanish at www.panther-program.org/skin-solutions. In some instances, using an implanted sensor can help avoid skin reactions in those sensitive to tape (100,101).

Substances and Factors Affecting Continuous Glucose Monitoring Accuracy

Sensor interference due to several medications/substances is a known potential source of CGM sensor measurement errors (Table 7.4). While several of these substances have been reported in the various CGM brands’ user manuals, additional interferences have been discovered after the market release of these products. Hydroxyurea, used for myeloproliferative disorders and hematologic conditions, is one of the most recently identified interfering substances that cause a temporary increase in sensor glucose values discrepant from actual glucose values (102,103). Similarly, substances such as mannitol and sorbitol, when administered intravenously or as a component of peritoneal dialysis solution, may increase blood mannitol or sorbitol concentrations and cause falsely elevated readings of sensor glucose (104). Therefore, it is crucial to routinely review the medications and supplements used by the person with diabetes to

Table 7.4—Continuous glucose monitoring device interfering substances

Medication	Systems affected	Effect
Acetaminophen >4 g/day Any dose	Dexcom G6, Dexcom G7 Medtronic Guardian 4	Higher sensor readings than actual glucose Higher sensor readings than actual glucose
Ascorbic acid (vitamin C), >500 mg/day	FreeStyle Libre 2, FreeStyle Libre 3	Higher sensor readings than actual glucose
Ascorbic acid (vitamin C), >1,000 mg/day	FreeStyle Libre 2 Plus, FreeStyle Libre 3 Plus	Higher sensor readings than actual glucose
Hydroxyurea	Dexcom G6, Dexcom G7, Medtronic Guardian 4	Higher sensor readings than actual glucose
Mannitol (intravenously or as peritoneal dialysis solution)	Senseonics Eversense365	Higher sensor readings than actual glucose
Sorbitol (intravenously or as peritoneal dialysis solution)	Senseonics Eversense365	Higher sensor readings than actual glucose

identify possible interfering substances and advise them accordingly on the need to use additional BGM if sensor values are unreliable due to these substances.

INSULIN DELIVERY

Insulin Syringes and Pens

Recommendations

7.21 For people with insulin-requiring diabetes on multiple daily injections (MDI), insulin pens are preferred in most cases. Still, insulin syringes may be used for insulin delivery considering individual and caregiver preference, insulin type, availability in vials, dosing therapy, cost, and self-management capabilities. **C**

7.22 Insulin pens or insulin injection aids are recommended for people with dexterity issues or vision impairment or when decided by shared decision making to facilitate the accurate dosing and administration of insulin. **C**

7.23 Offer connected insulin pens for people with diabetes taking multiple daily insulin injections when appropriate. **B**

7.24 FDA-approved insulin dose calculators/decision support systems may be helpful for calculating insulin doses. **B**

Injecting insulin with a syringe or pen (105–111) is the insulin delivery method used by most people with diabetes (112), although inhaled insulin is also available. Others use insulin pumps or AID devices (see INSULIN PUMPS AND AUTOMATED INSULIN DELIVERY SYSTEMS, below). For people with diabetes who use insulin, insulin syringes and pens both can deliver insulin safely and effectively for the achievement of glycemic goals. Individual preferences, cost, insulin type, dosing therapy, and self-management capabilities should be considered when choosing among delivery systems. Trials with insulin pens generally show equivalence or small improvements in glycemic outcomes compared with using a vial and syringe. Many individuals with diabetes prefer using a pen because of its simplicity and convenience. It is important to note that while many insulin types are available for purchase as either pens or vials, others may be available in only one form or the other, and there may be significant cost differences between pens and vials (see **Table 9.4** for a

list of insulin product costs with dosage forms). Insulin pens may allow people with vision impairment or dexterity issues to dose insulin accurately (113–115), and insulin injection aids are also available to help with these issues. (For a helpful list of injection aids, see diabetes.org/living-with-diabetes/treatment-care/diabetes-technology-guide). Inhaled technosphere insulin can be useful for people with diabetes, providing an alternative method of insulin delivery with very fast onset of action.

The most common syringe sizes are 1 mL, 0.5 mL, and 0.3 mL, allowing doses of up to 100 units, 50 units, and 30 units, respectively, of U-100 insulin. Some 0.3-mL syringes have half-unit markings, whereas other syringes have markings in 1- to 2-unit increments. In a few parts of the world, insulin syringes still have U-80 and U-40 markings for older insulin concentrations and veterinary insulin, and U-500 syringes are available for the use of U-500 insulin. Syringes are generally used once but may be reused by the same individual in resource-limited settings with appropriate storage and cleansing (115).

Insulin pens offer added convenience by combining the vial and syringe into a single device. Insulin pens, allowing push-button injections, come as disposable pens with prefilled cartridges or reusable insulin pens with replaceable insulin cartridges. Pens vary with respect to dosing increment and minimal dose, ranging from half-unit doses to 2-unit dose increments, with the latter available in U-200 insulin pens. U-500 pens come in 5-unit dose increments. Some reusable pens include a memory function, which can recall dose amounts and timing. Insulin pens, once started, can be kept in use for variable durations, based on the type of insulin, usually for 28 days, ranging from 14 to 56 days. Needle thickness (gauge) and length are other considerations. Needle gauges range from 22 to 34, with a higher gauge indicating a thinner needle. A thicker needle can give a dose of insulin more quickly, while a thinner needle may cause less pain. Needle length ranges from 4 to 12.7 mm, with some evidence suggesting that shorter needles (4–5 mm) lower the risk of intramuscular injection with erratic absorption and possibly the development of lipohypertrophy. When reused, needles may be duller and thus injections may be more painful. Proper insulin injection technique is a

requisite for receiving the full dose of insulin with each injection. Concerns with technique and use of the proper technique are outlined in section 9, “Pharmacologic Approaches to Glycemic Treatment.”

Connected insulin pens are insulin pens with the capacity to record and/or transmit insulin dose data. Insulin pen caps are also available and are placed on existing insulin pens and may assist with calculating insulin doses and by providing a memory function. Some connected insulin pens and pen caps can be programmed to calculate insulin doses, can be synced with select CGM systems, and can provide downloadable data reports. These pens and pen caps are useful to people with diabetes for real-time insulin dosing and allow clinicians to retrospectively review the insulin delivery times and, in some cases, doses and glucose data to make informed insulin dose adjustments (116). A quantitative study showed that people with diabetes preferred connected pens because of their ability to log insulin doses and glucose levels automatically (116). In a multicenter RCT in people with type 1 diabetes, the use of an insulin pen cap was associated with improved glycemic outcomes at 6 weeks in the insulin cap group, with an increase in TIR and decrease in GMI and TAR (117). A systematic review of connected insulin pens or pen caps showed improvement of glucose outcomes whether as A1C reduction, TIR increase, or hypoglycemia reduction (118). A recent real-world study with multinational data collected from 3,954 adults with diabetes using a connected pen and CGM validated the fact that treatment engagement with a connected insulin pen is positively associated with glycemic outcomes. On the other hand, missing as little as two basal doses or four bolus insulin doses over a 14-day period would be associated with a clinically relevant decrease in TIR of $\geq 5\%$ (119).

Bolus calculators have been developed to aid dosing decisions (120–125). These systems are subject to FDA clearance to ensure safety and efficacy in terms of algorithms used and subsequent dosing recommendations. People interested in using these systems should be encouraged to use those that are FDA approved. Health care professional input and education can be helpful for setting the initial dosing calculations

with ongoing follow-up for adjustments as needed.

Insulin Pumps and Automated Insulin Delivery Systems

Recommendations

7.25a AID systems are the preferred insulin delivery method over MDI, CSII, and sensor-augmented pumps in people with type 1 diabetes, **A** adults with type 2 diabetes, **A** children and adolescents with type 2 diabetes, **E** and those with other forms of insulin-deficient diabetes. **B, C, D, E** Choice of an AID system should be made based on the individual's circumstances, preferences, and needs. **E**

7.25b Consider AID systems for select people with type 2 diabetes treated with basal insulin not achieving individualized glycemic goals. **B** Choice of an AID system should be made based on the individual's circumstances, preferences, and needs. **E**

7.26 Individuals with diabetes who have been using CSII and/or AID should have continued access across third-party payors. **E**

Stand-alone Insulin Pumps and AID Systems

Insulin pumps have been available in the U.S. for over 40 years. These devices deliver rapid-acting insulin throughout the day to manage glucose levels. Most insulin pumps use tubing to deliver insulin through a cannula, while a few attach directly to the skin without tubing (pods or patch pumps), and these systems have been approved for use in type 1 and type 2 diabetes. AID systems, which can adjust insulin delivery rates based on sensor glucose values, are preferred over nonautomated pumps and MDI in people with type 1 diabetes and have largely replaced the use of nonintegrated or standard insulin pumps. Recently, three AID systems were approved for use by people with type 2 diabetes.

Historically, studies that compared MDI with insulin pump therapy were relatively small and of short duration. However, a systematic review and meta-analysis concluded that pump therapy in people with type 1 diabetes has modest advantages for lowering A1C levels (-0.30% [95% CI -0.58 to -0.02]) and for reducing severe hypoglycemia rates in children,

adolescents, and adults (126). Real-world data on insulin pump use in individuals with type 1 diabetes show benefits in A1C levels and hypoglycemia reductions as well as total daily insulin dose reduction (127).

Although AID systems have been shown to improve outcomes compared with MDI and insulin pump therapy alone, the choice of insulin delivery system is up to the individual along with support from their diabetes care team (126). Thus, the choice of insulin delivery method is based on the characteristics of the person with diabetes and which method is most likely to benefit them. Diabetes Wise (diabeteswise.org/), for individuals with diabetes, Diabetes Wise Pro (pro.diabeteswise.org/), for health care professionals, and the PANTHER Program (pantherprogram.org/device-comparison-chart) have helpful websites to assist health care professionals and people with diabetes in choosing diabetes devices based on their individual needs and preferences.

Adoption of AID and pump therapy in the U.S. shows geographical variations, which may be related to health care professional preference or center characteristics (128) and socioeconomic status, as pump therapy is more common in individuals of higher socioeconomic status, as reflected by private health insurance, family income, and education (128). Given the barriers to optimal diabetes care observed in disadvantaged groups (129), addressing the differences in access to insulin pumps and other diabetes technologies may lower health disparities (130).

AID systems preferentially, as well as insulin pump therapy alone or along with CGM, can be successfully started at the time of diagnosis (131) or at any point when they are needed in the course of an individual's diabetes. These devices can be safely used in adults (including older adults), adolescents, and children (15,16). Practical aspects of pump therapy initiation include assessment of readiness of the person with diabetes and their caregivers, if applicable. There is no consensus on which factors are most predictive of success with insulin pump or AID therapy (132). However, an understanding of the technical features of the system and how to troubleshoot if problems are encountered is necessary for safe use.

Complications and Challenges of Infusing Insulin

Complications of infused insulin can be caused by issues with infusion sets (dislodgement and occlusion), which put individuals at risk for ketosis and DKA and thus must be recognized and managed early (133). Other pump skin issues include lipohypertrophy or, less frequently, lipoatrophy (134) and pump site infection. Discontinuation of pump therapy is relatively uncommon today; the frequency has decreased over the past few decades, and its causes have changed (135). Current reasons for discontinuation are problems with cost or wearability, loss of insurance, dislike of the pump, suboptimal glycemic outcomes, or psychosocial considerations (e.g., anxiety or depression) (130). Common barriers to pump therapy adoption in children and adolescents are concerns regarding the physical interference of the device, discomfort with the idea of having a device on the body, therapeutic effectiveness, and financial burden (136–138).

Description of AID Systems

AID systems consist of mainly three components: an insulin pump, a CGM system, and an algorithm that determines insulin delivery. Based on the model and brand of currently FDA-approved and -cleared AID systems, the algorithm can be hosted in the pump body, in an insulin pod, or on a phone app. All AID systems on the market today integrate with one or more CGM systems and adjust insulin delivery either by modulating the preprogrammed basal rates or by replacing the basal rates with microboluses or microdoses of insulin every 5 min.

The modulation of insulin delivery is done by increasing, decreasing, or pausing insulin based on the CGM feedback, the predicted direction of the glucose levels, and the speed with which the glucose levels are changing. Different AID systems modulate insulin based on predicted glucose levels at various times, most commonly 30 min or 1 h. Currently available AID systems have either fixed glucose targets or adjustable glucose targets, ranging from 87 to 180 mg/dL (4.8 to 10 mmol/L), depending on the system. Glucose targets are generally set up for 24 h but can also be adjusted in some systems with up to eight segments per

day. All current AID systems provide automated correction doses, whether embedded in the microdose adjustments every 5 min or by providing additional correction boluses with doses that are dependent on the various types of algorithms, with variable frequency and threshold glucose based on the type of control algorithm. Most AID systems can be used in manual mode, although this is generally not recommended, as the benefits of CGM modulation may be partially or totally lost. However, use of AID in manual mode may be necessary in some circumstances, therefore it is important to review and reassess manual-mode settings periodically. Current AID systems still require manual entry of carbohydrates for meal announcements or qualitative meal estimation announcements to calculate prandial doses.

Adjustments for physical activity are available in most AID systems currently on the market. These can be programmed in various time increments. In general, the glucose target is raised to prespecified levels based on AID systems, and these are often accompanied by more conservative insulin delivery to reduce the risk of hypoglycemia in the setting of increased insulin sensitivity other than physical activity, such as prolonged fasting or NPO status for procedures. Of note, some systems may still give autocorrection boluses if the glucose levels rise above a certain threshold even while the exercise/activity mode has been enabled. Details on the available AID systems and their features can be found at pantherprogram.org/device-type.

AID systems have largely replaced other methods of continuous subcutaneous insulin delivery due to the advantages they offer in insulin modulation to adjust insulin doses and minimize hypoglycemia and hyperglycemia.

Partial Closed-Loop Systems

Sensor-augmented pumps (SAPs) were the precursors of the currently used AID systems. They are discussed here more for historical perspective because they are no longer commonly used, having been replaced by more fully functional AID systems. SAPs consist of three components: an insulin pump, a CGM system, and an algorithm that automates insulin suspension when glucose is low or is predicted to go low within the next 30 min. Predictive low-glucose suspend systems have been shown to reduce time spent with glucose <70 mg/dL

(<3.9 mmol/L) without rebound hyperglycemia during a 6-week randomized crossover trial in people with type 1 diabetes of various ages (139). Similar results were seen in additional studies in adults and children with reduction of hypoglycemia (140–142).

Data from Pivotal Trials

All currently FDA-approved and -cleared AID systems were tested for safety and efficacy in their pivotal trials in children, adolescents, and adults with type 1 diabetes (143–145) as well as in adults with type 2 diabetes (146,147). These studies were conducted by following a variety of methodologies, including observational studies as well as RCTs. Regardless of the study design, all AID system pivotal trials that examined individuals 2 years old or older, including older adults, have consistently demonstrated superiority to either standard insulin delivery or SAP and/or usual care (for the randomized trials), with consistent improvement in A1C, increase in TIR, especially overnight, as well as reduction of time spent in hypoglycemia (in the studies in people with type 1 diabetes). The greatest improvements were seen with AID when used in individuals with the highest baseline A1C or lowest TIR (148). These systems may also lower the risk of exercise-related hypoglycemia (149) and have been shown to have psychosocial benefits (150–155). A review of the literature on the health and economic value of AID systems found that AID systems are cost-effective (156). AID is the standard of care for people with type 1 diabetes who are capable of safely using the device and should be the preferred method of insulin delivery in these individuals. The decision to use AID systems should be made based on the preference of the person with diabetes (and/or caregivers) and the availability of resources to provide necessary training, access, and support for their use.

Data From Real-world Studies

Data from real-world studies on AID systems have become available and continue to increase rapidly. These studies include large numbers of users, at times even 30-fold higher than the number of people studied in AID pivotal trials (157). It is important to emphasize that for some AID systems all data are automatically collected to the database (158), whereas for other systems data

are collected based on voluntary sharing to the database by AID users. A recent systematic review of AID real-world studies, with 20 studies representing 171,209 individuals, substantiated the results observed in the pivotal trials and have confirmed the clinical benefits of AID systems in people with type 1 diabetes. Newer systems have shown increased time spent in automation, and the real-world studies have retrospectively analyzed longer duration of system use compared with their respective pivotal trials, with most analyses occurring for more than 6 months and an average duration of 9 months (157). Benefits include improvement in A1C levels, TIR, and other glucometrics as well as psychosocial benefits (159–164).

Finally, real-world data showed that AID systems provide the same glycemic benefits to Medicare and Medicaid beneficiaries with type 1 and type 2 diabetes, emphasizing that access to this technology should be made available regardless of A1C levels and should be based on the individual's needs (165).

Automated Insulin Delivery Systems in Pregnancy

The use of AID systems in diabetes and pregnancy presents particular challenges, as the current FDA-approved AID systems (except for one that has been FDA approved but is not yet commercially available) have glucose goals that are not pregnancy specific and do not have algorithms designed to achieve pregnancy-specific glucose goals. Initiating or continuing AID systems during pregnancy needs to be assessed carefully. Selected individuals with type 1 diabetes should be evaluated as potential candidates for AID systems in the setting of expert guidance. The details on the use of AID systems in pregnancy are discussed in section 15, "Management of Diabetes in Pregnancy."

Insulin Pumps and Automated Insulin Delivery Systems in People With Type 2 and Other Types of Diabetes

Additional insulin delivery options in people with type 2 diabetes include disposable patch-like devices, which provide either a continuous subcutaneous insulin infusion of rapid-acting insulin (basal) with bolus insulin in 2-unit increments at the press of a button or bolus insulin only, delivered in 2-unit increments, used in

conjunction with basal insulin injections (166–169). Use of a technology as a means of insulin delivery is an individual choice for people with diabetes and should be considered an option in those who are capable of safely using the device.

Open-Source Automated Insulin Dosing

Recommendation

7.27 Support and provide diabetes management advice to people with diabetes who choose to use an open-source AID system. **B**

Open-source automated insulin dosing (OS-AID) algorithms provide the precise code that governs their operation, so health care professionals and people with diabetes can have a more complete understanding of risks and benefits of their use (170). Any commercial entity could provide the source code for their interoperable automated glycemic controller, but most choose not to. OS-AID algorithms are largely designed, maintained, and curated by people with diabetes and their caregivers. Many people with diabetes use these algorithms with cleared CGM systems and insulin pump components. The information on how to set up and manage these systems is freely available online.

OS-AID is the preferred term when referring to any open-source system (commercial or otherwise). It is important to note that the term “DIY” is not reflective of any aspect of these community-driven systems. No individual person has written all the code for these algorithms, and a large percentage of users do not build the software themselves (171). There are two main available algorithms, the OpenAPS algorithm and the Loop algorithm, which have been implemented on a variety of platforms.

The OpenAPS heuristic algorithm (implemented on a system on a chip in OpenAPS, Android smartphones as AndroidAPS, and iPhone as iAPS/Trio) is supported by large real-world studies (172) and a multicenter RCT (173). The OpenAPS algorithm also supports unannounced meals. In a single-center study of adolescents with type 1 diabetes randomized to AndroidAPS with quantitative carbohydrate announcements, qualitative announcements,

and no announcements, TIR was preserved across groups (174).

Loop, an open-source model predictive control algorithm, is implemented on iPhones as an app. Prospective real-world data from 558 adults and children with type 1 diabetes on this system (171) was used to support the FDA clearance of a variant called Tidepool Loop (175), now implemented commercially.

Both the Loop and OpenAPS algorithms offer direct management of algorithm aggressiveness through conventional pump settings. Therefore, it is advisable that health care professionals understand and offer support in tuning settings for these safe and effective technologies (170). This may include, for example, the adjustment of basal rates, insulin-to-carbohydrate ratios, or insulin sensitivity factors. As with any AID system, a backup insulin treatment plan is advisable.

Digital Health Technology

Recommendation

7.28 Consider combining technology (CGM, insulin pump, and/or diabetes apps) with online or virtual licensed coaching to improve glycemic outcomes in individuals with diabetes or prediabetes. **B**

Increasingly, people are turning to the internet for advice, coaching, connection, and health care. Diabetes, partly because it is both common and numeric, lends itself to the development of apps and online programs. Recommendations for developing and implementing a digital diabetes clinic have been published (176). The FDA approves and monitors clinically validated, digital, and usually online health technologies intended to treat a medical or psychological condition; these are known as digital therapeutics, or “digiceuticals” (fda.gov/medical-devices/digital-health-center-excellence/device-software-functions-including-mobile-medical-applications) (177). Other applications, such as those that assist in displaying or storing data, encourage a healthy lifestyle or provide limited clinical data support. Therefore, it is possible to find apps that have been fully reviewed and approved by the FDA and others designed and promoted by people with relatively little skill or knowledge in the clinical treatment of diabetes. There are insufficient data to

provide recommendations for specific apps for diabetes management, education, and support in the absence of RCTs and validation of apps unless they are FDA cleared.

An area of particular importance is that of online privacy and security. Established cloud-based data aggregator programs, such as Tidepool, Glooko, and others, have been developed with appropriate data security features and are compliant with the U.S. Health Insurance Portability and Accountability Act of 1996. These programs can help monitor people with diabetes and provide access to their health care teams (178). Consumers should read the policy regarding data privacy and sharing before entering data into an application and learn how they can manage the way their data will be used (some programs offer the ability to share more or less information, such as being part of a registry or data repository or not).

Many online programs offer lifestyle counseling to achieve weight loss and increased physical activity (179). Many include a health coach and can create small groups of similar participants on social networks. Some programs aim to treat prediabetes and prevent progression to diabetes, often following the model of the Diabetes Prevention Program (180,181). Others assist in improving diabetes outcomes by remotely monitoring clinical data (for instance, wireless monitoring of glucose levels, weight, or blood pressure) and providing feedback and coaching (182–187). There are text messaging approaches that tie into a variety of different types of lifestyle and treatment programs, which vary in terms of their effectiveness (188,189). There are limited RCT data for many of these interventions, and long-term follow-up is lacking. However, in a real-world observational study of individuals with type 2 diabetes treated with basal insulin, noninsulin glucose-lowering medications, or no medications, the use of a digital health solution and rtCGM resulted in reductions of GMI and TAR >180 mg/dL (>10 mmol/L) and >250 mg/dL (>13.9 mmol/L) as well as an increase in TIR by 15% and participation in a least one engagement activity per week (190). Therefore, even with limited data, for an individual with diabetes, opting in to one of these programs can

be helpful in providing support and, for many, is an attractive option.

Inpatient Care

Recommendations

7.29 In people with diabetes wearing personal CGM, the use of CGM should be continued when clinically appropriate during hospitalization, with confirmatory point-of-care glucose measurements for insulin dosing and hypoglycemia assessment and treatment under an institutional protocol. **B**
7.30 Continue use of insulin pump or AID in people with diabetes who are hospitalized when clinically appropriate. This is contingent upon availability of necessary supplies, resources, training, ongoing competency assessments, and implementation of institutional diabetes technology protocols. **C**

For details on technology use in the inpatient setting, see section 16, “Diabetes Care in the Hospital.”

The Future

The pace of development in diabetes technology is extremely rapid. New approaches and tools are available each year. It is difficult for research to keep up with these advances, because newer versions of the devices and digital solutions are already on the market by the time a study is completed. The most important consideration in all these systems is the person with diabetes. Technology selection must be appropriate and safe for the individual. Simply having a device or application does not change outcomes unless the human being engages with it appropriately to create positive health benefits. This underscores the need for the health care team to assist people with diabetes in device and program selection and to support their use through ongoing education and training. Expectations must be tempered by reality—we do not yet have technology that completely eliminates the self-care tasks necessary for managing diabetes, but the tools described in this section can make it easier to manage if disseminated equitably.

References

- Broos B, Charleer S, Bolsens N, et al. Diabetes knowledge and metabolic control in type 1 diabetes starting with continuous glucose monitoring: FUTURE-PEAK. *J Clin Endocrinol Metab* 2021;106:e3037–e3048
- Yoo JH, Kim G, Lee HJ, Sim KH, Jin S-M, Kim JH. Effect of structured individualized education on continuous glucose monitoring use in poorly controlled patients with type 1 diabetes: a randomized controlled trial. *Diabetes Res Clin Pract* 2022;184:109209
- Champakanath A, Akturk HK, Alonso GT, Snell-Bergeon JK, Shah VN. Continuous glucose monitoring initiation within first year of type 1 diabetes diagnosis is associated with improved glycemic outcomes: 7-year follow-up study. *Diabetes Care* 2022;45:750–753
- Patton SR, Noser AE, Youngkin EM, Majidi S, Clements MA. Early initiation of diabetes devices relates to improved glycemic control in children with recent-onset type 1 diabetes mellitus. *Diabetes Technol Ther* 2019;21:379–384
- Prahalad P, Ding VY, Zaharieva DP, et al. Teamwork, targets, technology, and tight control in newly diagnosed type 1 diabetes: the Pilot 4T Study. *J Clin Endocrinol Metab* 2022;107:998–1008
- Addala A, Maahs DM, Scheinker D, Chertow S, Leverenz B, Prahalad P. Uninterrupted continuous glucose monitoring access is associated with a decrease in HbA1c in youth with type 1 diabetes and public insurance. *Pediatr Diabetes* 2020;21:1301–1309
- Tanenbaum ML, Zaharieva DP, Addala A, et al. “I was ready for it at the beginning”: parent experiences with early introduction of continuous glucose monitoring following their child’s type 1 diabetes diagnosis. *Diabet Med* 2021;38:e14567
- Prahalad P, Ding VY, Zaharieva DP, et al. Sustained HbA1c improvements over 36 months in youth in the Teamwork, Targets, Technology, and Tight Control (4T) study. *J Clin Endo Met*. 10 July 2025 [Epub ahead of print]. DOI: 10.1210/clinem/dgaf397
- Maizel J, Haller MJ, Maahs DM, et al. Peer mentoring improves diabetes technology use and reduces diabetes distress among underserved communities: outcomes of a pilot diabetes support coach intervention. *J Diabetes Res* 2025;2025:1970247
- Weinstock RS, Raghinaru D, Gal RL, et al. Older adults benefit from virtual support for continuous glucose monitor use but require longer visits. *J Diabetes Sci Technol*. 2 November 2024 [Epub ahead of print]. DOI: 10.1177/19322968241294250
- Meighan S, Douvas JL, Rearson A, Squaresky R, Kelly A, Marks BE. The type of patient training does not impact outcomes in the first 90 days of automated insulin delivery use. *Diabetes Technol Ther* 2024;26:773–779
- Patil SP, Albanese-O’Neill A, Yehl K, Seley JJ, Hughes AS. Professional competencies for diabetes technology use in the care setting. *Sci Diabetes Self Manag Care* 2022;48:437–445
- Phillip M, Nimri R, Bergenstal RM, et al. Consensus recommendations for the use of automated insulin delivery technologies in clinical practice. *Endocr Rev* 2023;44:254–280
- Anderson JE, Greene MA, Griffin JW, Jr, et al.; American Diabetes Association. Diabetes and employment. *Diabetes Care* 2014;37(Suppl. 1):S112–S117
- Boughton CK, Allen JM, Ware J, et al. The effect of closed-loop glucose control on C-peptide secretion in youth with newly diagnosed type 1 diabetes: the CLOuD RCT. *Efficacy Mech Eval* 2024;1–75
- Vakharia M, Lyons SK, Buckingham D, et al. Initiating insulin pumps in youth with new-onset type 1 diabetes: a quality improvement initiative. *Pediatr Qual Saf* 2025;10:e803
- Aleppo G, Beck RW, Bailey R, et al.; Type 2 Diabetes Basal Insulin Users: The Mobile Study (MOBILE) Study Group. The effect of discontinuing continuous glucose monitoring in adults with type 2 diabetes treated with basal insulin. *Diabetes Care* 2021;44:2729–2737
- Aronson R, Abitbol A, Bajaj HS, et al. Continuous glucose monitoring in noninsulin-treated type 2 diabetes: a critical review of reported trials with an updated systematic review and meta-analysis of randomised controlled trials. *Diabetes Obes Metab* 2025;27:6220–6242
- Karakus KE, Akturk HK, Alonso GT, Snell-Bergeon JK, Shah VN. Association between diabetes technology use and glycemic outcomes in adults with type 1 diabetes over a decade. *Diabetes Care* 2023;46:1646–1651
- Vigersky RA, Huang S, Cordero TL, et al.; OpT2mise Study Group. Improved HbA1c, total daily insulin dose, and treatment satisfaction with insulin pump therapy compared to multiple daily insulin injections in patients with type 2 diabetes irrespective of baseline C-peptide levels. *Endocr Pract* 2018;24:446–452
- Gill M, Chhabra H, Shah M, Zhu C, Grunberger G. C-peptide and beta-cell autoantibody testing prior to initiating continuous subcutaneous insulin infusion pump therapy did not improve utilization or medical costs among older adults with diabetes mellitus. *Endocr Pract* 2018;24:634–645
- Aleppo G, Parkin CG, Carlson AL, et al. Lost in translation: a disconnect between the science and medicare coverage criteria for continuous subcutaneous insulin infusion. *Diabetes Technol Ther* 2021;23:715–725
- Argento NB, Liu J, Hughes AS, McAuliffe-Fogarty AH. Impact of Medicare continuous subcutaneous insulin infusion policies in patients with type 1 diabetes. *J Diabetes Sci Technol* 2020;14:257–261
- Mullen DM, Bergenstal RM, Johnson M, et al. AGP reports for glucose and insulin devices qualitative study: what patients and clinicians want. *Sci Diabetes Self Manag Care* 2025;51:333–344
- Simonson GD, Criego AB, Battelino T, et al. Expert panel recommendations for a standardized ambulatory glucose profile report for connected insulin pens. *Diabetes Technol Ther* 2024;26:814–822
- Johnson ML, Martens TW, Criego AB, Carlson AL, Simonson GD, Bergenstal RM. Utilizing the ambulatory glucose profile to standardize and implement continuous glucose monitoring in clinical practice. *Diabetes Technol Ther* 2019;21:S217–S225
- Karakus KE, Snell-Bergeon JK, Akturk HK. Comparison of computational statistical packages for the analysis of continuous glucose monitoring data with a reference software, “Ambulatory Glucose Profile,” in type 1 diabetes. *Diabetes Technol Ther* 2025;27:202–208
- Diabetes Control and Complications Trial (DCCT)/Epidemiology of Diabetes Interventions and Complications (EDIC) Study Research Group. Intensive diabetes treatment and cardiovascular outcomes in type 1 diabetes: the DCCT/EDIC

- study 30-year follow-up. *Diabetes Care* 2016;39:686–693
29. Pleus S, Baumstark A, Jendrike N, et al. System accuracy evaluation of 18 CE-marked current-generation blood glucose monitoring systems based on EN ISO 15197:2015. *BMJ Open Diabetes Res Care* 2020;8:e001067
 30. Grady M, Lamps G, Shemain A, Cameron H, Murray L. Clinical evaluation of a new, lower pain, one touch lancing device for people with diabetes: virtually pain-free testing and improved comfort compared to current lancing systems. *J Diabetes Sci Technol* 2021;15:53–59
 31. Heinemann L, Drossel D, Freckmann G, Kulzer B. Usability of medical devices for patients with diabetes who are visually impaired or blind. *J Diabetes Sci Technol* 2016;10:1382–1387
 32. Harrison B, Brown D. Accuracy of a blood glucose monitoring system that recognizes insufficient sample blood volume and allows application of more blood to the same test strip. *Expert Rev Med Devices* 2020;17:75–82
 33. Karter AJ, Ackerson LM, Darbinian JA, et al. Self-monitoring of blood glucose levels and glycemic control: the Northern California Kaiser Permanente Diabetes registry. *Am J Med* 2001;111:1–9
 34. Miller KM, Beck RW, Bergenstal RM, et al.; T1D Exchange Clinic Network. Evidence of a strong association between frequency of self-monitoring of blood glucose and hemoglobin A1c levels in T1D exchange clinic registry participants. *Diabetes Care* 2013;36:2009–2014
 35. Grant RW, Huang ES, Wexler DJ, et al. Patients who self-monitor blood glucose and their unused testing results. *Am J Manag Care* 2015;21:e119–e129
 36. Katz LB, Stewart L, Guthrie B, Cameron H. Patient satisfaction with a new, high accuracy blood glucose meter that provides personalized guidance, insight, and encouragement. *J Diabetes Sci Technol* 2020;14:318–323
 37. Shaw RJ, Yang Q, Barnes A, et al. Self-monitoring diabetes with multiple mobile health devices. *J Am Med Inform Assoc* 2020;27:667–676
 38. Gellad WF, Zhao X, Thorpe CT, Mor MK, Good CB, Fine MJ. Dual use of Department of Veterans Affairs and Medicare benefits and use of test strips in veterans with type 2 diabetes mellitus. *JAMA Intern Med* 2015;175:26–34
 39. Ziegler R, Heidtmann B, Hilgard D, Hofer S, Rosenbauer J, Holl R; DPV-Wiss-Initiative. Frequency of SMBG correlates with HbA1c and acute complications in children and adolescents with type 1 diabetes. *Pediatr Diabetes* 2011;12:11–17
 40. Rosenstock J, Bajaj HS, Lingvay I, Heller SR. Clinical perspectives on the frequency of hypoglycemia in treat-to-target randomized controlled trials comparing basal insulin analogs in type 2 diabetes: a narrative review. *BMJ Open Diabetes Res Care* 2024;12:e003930
 41. Zou Y, Zhao S, Li G, Zhang C. The efficacy and frequency of self-monitoring of blood glucose in non-insulin-treated T2D patients: a systematic review and meta-analysis. *J Gen Intern Med* 2023;38:755–764
 42. Young LA, Buse JB, Weaver MA, et al.; Monitor Trial Group. Glucose self-monitoring in non-insulin-treated patients with type 2 diabetes in primary care settings: a randomized trial. *JAMA Intern Med* 2017;177:920–929
 43. Holmes-Truscott E, Baptista S, Ling M, et al. The impact of structured self-monitoring of blood glucose on clinical, behavioral, and psychosocial outcomes among adults with non-insulin-treated type 2 diabetes: a systematic review and meta-analysis. *Front Clin Diabetes Healthc* 2023;4:1177030
 44. Sai S, Urata M, Ogawa I. Evaluation of linearity and interference effect on SMBG and POCT devices, showing drastic high values, low values, or error messages. *J Diabetes Sci Technol* 2019;13:734–743
 45. Sacks DB, Arnold M, Bakris GL, et al. Guidelines and recommendations for laboratory analysis in the diagnosis and management of diabetes mellitus. *Diabetes Care* 2023;46:e151–e199
 46. Tamborlane WV, Beck RW, Bode BW, et al.; Juvenile Diabetes Research Foundation Continuous Glucose Monitoring Study Group. Continuous glucose monitoring and intensive treatment of type 1 diabetes. *N Engl J Med* 2008;359:1464–1476
 47. Tumminia A, Crimi S, Sciacca L, et al. Efficacy of real-time continuous glucose monitoring on glycaemic control and glucose variability in type 1 diabetic patients treated with either insulin pumps or multiple insulin injection therapy: a randomized controlled crossover trial. *Diabetes Metab Res Rev* 2015;31:61–68
 48. Hansen KW, Bibby BM. The frequency of intermittently scanned glucose and diurnal variation of glycemic metrics. *J Diabetes Sci Technol* 2022;16:1461–1465
 49. Urakami T, Yoshida K, Kuwabara R, et al. Frequent scanning using flash glucose monitoring contributes to better glycemic control in children and adolescents with type 1 diabetes. *J Diabetes Investig* 2022;13:185–190
 50. Laffel LM, Kanapka LG, Beck RW, et al.; CDE10. Effect of continuous glucose monitoring on glycemic control in adolescents and young adults with type 1 diabetes: a randomized clinical trial. *JAMA* 2020;323:2388–2396
 51. Strategies to Enhance New CGM Use in Early Childhood (SENCE) Study Group. A randomized clinical trial assessing continuous glucose monitoring (CGM) use with standardized education with or without a family behavioral intervention compared with fingerstick blood glucose monitoring in very young children with type 1 diabetes. *Diabetes Care* 2021;44:464–472
 52. New JP, Ajjan R, Pfeiffer AFH, Freckmann G. Continuous glucose monitoring in people with diabetes: the randomized controlled Glucose Level Awareness in Diabetes Study (GLADIS). *Diabet Med* 2015;32:609–617
 53. Gubitosi-Klug RA, Braffett BH, Bebu I, et al. Continuous glucose monitoring in adults with type 1 diabetes with 35 years duration from the DCCT/EDIC study. *Diabetes Care* 2022;45:659–665
 54. Sequeira PA, Montoya L, Ruelas V, et al. Continuous glucose monitoring pilot in low-income type 1 diabetes patients. *Diabetes Technol Ther* 2013;15:855–858
 55. Friedman JG, Coyne K, Aleppo G, Szmuliowicz ED. Beyond A1C: exploring continuous glucose monitoring metrics in managing diabetes. *Endocr Connect* 2023;12:e230085
 56. Teo E, Hassan N, Tam W, Koh S. Effectiveness of continuous glucose monitoring in maintaining glycaemic control among people with type 1 diabetes mellitus: a systematic review of randomised controlled trials and meta-analysis. *Diabetologia* 2022;65:604–619
 57. Pratley RE, Kanapka LG, Rickels MR, et al.; Wireless Innovation for Seniors With Diabetes Mellitus (WISDM) Study Group. Effect of continuous glucose monitoring on hypoglycemia in older adults with type 1 diabetes: a randomized clinical trial. *JAMA* 2020;323:2397–2406
 58. Miller KM, Kanapka LG, Rickels MR, et al. Benefit of continuous glucose monitoring in reducing hypoglycemia is sustained through 12 months of use among older adults with type 1 diabetes. *Diabetes Technol Ther* 2022;24:424–434
 59. Bao S, Bailey R, Calhoun P, Beck RW. Effectiveness of continuous glucose monitoring in older adults with type 2 diabetes treated with basal insulin. *Diabetes Technol Ther* 2022;24:299–306
 60. Van Name MA, Kanapka LG, DiMeglio LA, et al. Long-term continuous glucose monitor use in very young children with type 1 diabetes: one-year results from the SENCE study. *J Diabetes Sci Technol* 2023;17:976–987
 61. Martens T, Beck RW, Bailey R, et al.; MOBILE Study Group. Effect of continuous glucose monitoring on glycemic control in patients with type 2 diabetes treated with basal insulin: a randomized clinical trial. *JAMA* 2021;325:2262–2272
 62. Grace T, Salyer J. Use of real-time continuous glucose monitoring improves glycemic control and other clinical outcomes in type 2 diabetes patients treated with less intensive therapy. *Diabetes Technol Ther* 2022;24:26–31
 63. Ehrhardt NM, Chellappa M, Walker MS, Fonda SJ, Vigersky RA. The effect of real-time continuous glucose monitoring on glycemic control in patients with type 2 diabetes mellitus. *J Diabetes Sci Technol* 2011;5:668–675
 64. Price DA, Deng Q, Kipnes M, Beck SE. Episodic real-time CGM use in adults with type 2 diabetes: results of a pilot randomized controlled trial. *Diabetes Ther* 2021;12:2089–2099
 65. Jancev M, Vissers TACM, Visseren FLJ, et al. Continuous glucose monitoring in adults with type 2 diabetes: a systematic review and meta-analysis. *Diabetologia* 2024;67:798–810
 66. Ferreira ROM, Trevisan T, Pasqualotto E, et al. Continuous glucose monitoring systems in noninsulin-treated people with type 2 diabetes: a systematic review and meta-analysis of randomized controlled trials. *Diabetes Technol Ther* 2024;26:252–262
 67. Moon SJ, Kim K-S, Lee WJ, Lee MY, Vigersky R, Park C-Y. Efficacy of intermittent short-term use of a real-time continuous glucose monitoring system in non-insulin-treated patients with type 2 diabetes: a randomized controlled trial. *Diabetes Obes Metab* 2023;25:110–120
 68. Manfredo J, Lin T, Gupta R, et al. Short-term use of CGM in youth onset type 2 diabetes is associated with behavioral modifications. *Front Endocrinol (Lausanne)* 2023;14:1182260
 69. Chesser H, Srinivasan S, Puckett C, Gitelman SE, Wong JC. Real-time continuous glucose monitoring in adolescents and young adults with type 2 diabetes can improve quality of life. *J Diabetes Sci Technol* 2024;18:911–919
 70. Deshmukh H, Wilmut EG, Gregory R, et al. Effect of flash glucose monitoring on glycemic control, hypoglycemia, diabetes-related distress, and resource utilization in the Association of British Clinical Diabetologists (ABCD) nationwide audit. *Diabetes Care* 2020;43:2153–2160

71. Charleer S, Gillard P, Vandoorne E, Cammaerts K, Mathieu C, Casteels K. Intermittently scanned continuous glucose monitoring is associated with high satisfaction but increased HbA1c and weight in well-controlled youth with type 1 diabetes. *Pediatr Diabetes* 2020;21:1465–1474
72. Krakauer M, Botero JF, Lavalle-González FJ, Proietti A, Barbieri DE. A review of flash glucose monitoring in type 2 diabetes. *Diabetol Metab Syndr* 2021;13:42
73. Hohendorf J, Gumprecht J, Mysliwiec M, Zozulinska-Ziolkiewicz D, Malecki MT. Intermittently scanned continuous glucose monitoring data of polish patients from real-life conditions: more scanning and better glycemic control compared to worldwide data. *Diabetes Technol Ther* 2021;23: 577–585
74. Nathanson D, Svensson A-M, Miftaraj M, Franzén S, Bolinder J, Eeg-Olofsson K. Effect of flash glucose monitoring in adults with type 1 diabetes: a nationwide, longitudinal observational study of 14,372 flash users compared with 7691 glucose sensor naïve controls. *Diabetologia* 2021; 64:1595–1603
75. Roussel R, Riveline J-P, Vicaud E, et al. Important drop in rate of acute diabetes complications in people with type 1 or type 2 diabetes after initiation of flash glucose monitoring in France: the RELIEF study. *Diabetes Care* 2021;44:1368–1376
76. Riveline J-P, Roussel R, Vicaud E, et al. Reduced rate of acute diabetes events with flash glucose monitoring is sustained for 2 years after initiation: extended outcomes from the RELIEF study. *Diabetes Technol Ther* 2022;24:611–618
77. Miller E, Kerr MSD, Roberts GJ, Nabutovsky Y, Wright E. Flash CGM associated with event reduction in nonintensive diabetes therapy. *Am J Manag Care* 2021;27:e372–e377
78. Karter AJ, Parker MM, Moffet HH, Gilliam LK, Dlott R. Association of real-time continuous glucose monitoring with glycemic control and acute metabolic events among patients with insulin-treated diabetes. *JAMA* 2021;325:2273–2284
79. Layne JE, Jepson LH, Carite AM, Parkin CG, Bergenstal RM. Long-term improvements in glycemic control with Dexcom CGM use in adults with noninsulin-treated type 2 diabetes. *Diabetes Technol Ther* 2024;26:925–931
80. Akturk HK; American Diabetes Association Diabetes Technology Interest Group. Recent advances in diabetes technology and activities of the American Diabetes Association Diabetes Technology Interest Group. *Clin Diabetes* 2024; 42:316–321
81. Dexcom, Inc. *Dexcom G7 Continuous Glucose Monitoring System. Integrated Continuous Glucose Monitoring System, Factory Calibrated*. Dexcom, Inc. Accessed 14 August 2025. Available from <https://fda.report/PMN/K213919>
82. Abitbol A, Jain AB, Tsoukas MA, et al. Use of flash glucose monitoring is associated with HbA1c reduction in type 2 diabetes managed with basal insulin in Canada: a real-world prospective observational study. *Diab Vasc Dis Res* 2024;21: 14791641241253967
83. Heer RS, Lovegrove J, Welsh Z. Efficacy of flash glucose monitoring on HbA1c in type 2 diabetes: an individual patient data meta-analysis of real-world evidence. *Diabetes Res Clin Pract* 2025;219:111950
84. Ribeiro RT, Andrade R, Nascimento do Ó D, Lopes AF, Raposo JF. Impact of blinded retrospective continuous glucose monitoring on clinical decision making and glycemic control in persons with type 2 diabetes on insulin therapy. *Nutr Metab Cardiovasc Dis* 2021;31:1267–1275
85. Ziegler R, Heinemann L, Freckmann G, Schnell O, Hinzmann R, Kulzer B. Intermittent use of continuous glucose monitoring: expanding the clinical value of CGM. *J Diabetes Sci Technol* 2021;15:684–694
86. Wada E, Onoue T, Kobayashi T, et al. Flash glucose monitoring helps achieve better glycemic control than conventional self-monitoring of blood glucose in non-insulin-treated type 2 diabetes: a randomized controlled trial. *BMJ Open Diabetes Res Care* 2020;8:e001115
87. Nemlekar PM, Hannah KL, Norman GJ. Association between change in A1C and use of professional continuous glucose monitoring in adults with type 2 diabetes on noninsulin therapies: a real-world evidence study. *Clin Diabetes* 2023;41: 359–366
88. Sherrill CH, Houpt CT, Dixon EM, Richter SJ. Effect of pharmacist-driven professional continuous glucose monitoring in adults with uncontrolled diabetes. *J Manag Care Spec Pharm* 2020;26: 600–609
89. Fantasia KL, Stockman M-C, Ju Z, et al. Professional continuous glucose monitoring and endocrinology eConsult for adults with type 2 diabetes in primary care: results of a clinical pilot program. *J Clin Transl Endocrinol* 2021;24:100254
90. Simonson GD, Bergenstal RM, Johnson ML, Davidson JL, Martens TW. Effect of professional CGM (pCGM) on glucose management in type 2 diabetes patients in primary care. *J Diabetes Sci Technol* 2021;15:539–545
91. Pleus S, Ulbrich S, Zschornack E, Kamann S, Haug C, Freckmann G. Documentation of skin-related issues associated with continuous glucose monitoring use in the scientific literature. *Diabetes Technol Ther* 2019;21:538–545
92. Herman A, de Montjoye L, Baek M. Adverse cutaneous reaction to diabetic glucose sensors and insulin pumps: irritant contact dermatitis or allergic contact dermatitis? *Contact Dermatitis* 2020;83:25–30
93. Rigo RS, Levin LE, Belsito DV, Garzon MC, Gandica R, Williams KM. Cutaneous reactions to continuous glucose monitoring and continuous subcutaneous insulin infusion devices in type 1 diabetes mellitus. *J Diabetes Sci Technol* 2021;15: 786–791
94. Kamann S, Aerts O, Heinemann L. Further evidence of severe allergic contact dermatitis from isobornyl acrylate while using a continuous glucose monitoring system. *J Diabetes Sci Technol* 2018;12:630–633
95. Hyry HSI, Liippo JP, Virtanen HM. Allergic contact dermatitis caused by glucose sensors in type 1 diabetes patients. *Contact Dermatitis* 2019; 81:161–166
96. Asarani NAM, Reynolds AN, Boucher SE, de Bock M, Wheeler BJ. Cutaneous complications with continuous or flash glucose monitoring use: systematic review of trials and observational studies. *J Diabetes Sci Technol* 2020;14:328–337
97. Lombardo F, Salzano G, Crisafulli G, et al. Allergic contact dermatitis in pediatric patients with type 1 diabetes: an emerging issue. *Diabetes Res Clin Pract* 2020;162:108089
98. Oppel E, Kamann S, Heinemann L, Reichl F-X, Högg C. The implanted glucose monitoring system Eversense: an alternative for diabetes patients with isobornyl acrylate allergy. *Contact Dermatitis* 2020;82:101–104
99. Freckmann G, Buck S, Waldenmaier D, et al. Skin reaction report form: development and design of a standardized report form for skin reactions due to medical devices for diabetes management. *J Diabetes Sci Technol* 2021;15: 801–806
100. Deiss D, Irace C, Carlson G, Tweden KS, Kaufman FR. Real-world safety of an implantable continuous glucose sensor over multiple cycles of use: a post-market registry study. *Diabetes Technol Ther* 2020;22:48–52
101. Sanchez P, Ghosh-Dastidar S, Tweden KS, Kaufman FR. Real-world data from the first U.S. commercial users of an implantable continuous glucose sensor. *Diabetes Technol Ther* 2019;21: 677–681
102. Szmulowicz ED, Aleppo G. Interferent effect of hydroxyurea on continuous glucose monitoring. *Diabetes Care* 2021;44:e89–e90
103. Pfützner A, Jensch H, Cardinal C, Srikanthamoorthy G, Riehn E, Thomé N. Laboratory protocol and pilot results for dynamic interference testing of continuous glucose monitoring sensors. *J Diabetes Sci Technol* 2024;18:59–65
104. U.S. FDA. Summary of safety and effectiveness data (SSED). Continuous glucose monitor (CGM), implanted, adjunctive use 2018. Accessed 14 August 2025. Available from https://www.accessdata.fda.gov/cdrh_docs/pdf16/P160048B.pdf
105. Machry RV, Cipriani GF, Pedrosa HU, et al. Pens versus syringes to deliver insulin among elderly patients with type 2 diabetes: a randomized controlled clinical trial. *Diabetol Metab Syndr* 2021;13:64
106. Korytkowski M, Bell D, Jacobsen C, Suwannasari R; FlexPen Study Team. A multicenter, randomized, open-label, comparative, two-period crossover trial of preference, efficacy, and safety profiles of a prefilled, disposable pen and conventional vial/syringe for insulin injection in patients with type 1 or 2 diabetes mellitus. *Clin Ther* 2003;25:2836–2848
107. Singh R, Samuel C, Jacob JJ. A comparison of insulin pen devices and disposable plastic syringes - simplicity, safety, convenience and cost differences. *Eur Endocrinol* 2018;14:47–51
108. Frid AH, Kreugel G, Grassi G, et al. New insulin delivery recommendations. *Mayo Clin Proc* 2016;91:1231–1255
109. Slabaugh SL, Bouchard JR, Li Y, Baltz JC, Meah YA, Moretz DC. Characteristics relating to adherence and persistence to basal insulin regimens among elderly insulin-naïve patients with type 2 diabetes: pre-filled pens versus vials/syringes. *Adv Ther* 2015;32:1206–1221
110. Ahmann A, Szeinbach SL, Gill J, Traylor L, Garg SK. Comparing patient preferences and healthcare provider recommendations with the pen versus vial-and-syringe insulin delivery in patients with type 2 diabetes. *Diabetes Technol Ther* 2014;16:76–83
111. Asche CV, Luo W, Aagren M. Differences in rates of hypoglycemia and health care costs in patients treated with insulin aspart in pens versus vials. *Curr Med Res Opin* 2013;29:1287–1296
112. Hanas R, de Beaufort C, Hoey H, Anderson B. Insulin delivery by injection in children and adolescents with diabetes. *Pediatr Diabetes* 2011; 12:518–526

113. Pfützner A, Schipper C, Niemeyer M, et al. Comparison of patient preference for two insulin injection pen devices in relation to patient dexterity skills. *J Diabetes Sci Technol* 2012;6:910–916
114. Maltese G, McAuley SA, Trawley S, Sinclair AJ. Ageing well with diabetes: the role of technology. *Diabetologia* 2024;67:2085–2102
115. Zabaleta-Del-Olmo E, Vlachos B, Jodar-Fernández L, et al. Safety of the reuse of needles for subcutaneous insulin injection: a systematic review and meta-analysis. *Int J Nurs Stud* 2016;60:121–132
116. Seo J, Heidenreich S, Aldalooj E, et al. Patients' preferences for connected insulin pens: a discrete choice experiment among patients with type 1 and type 2 diabetes. *Patient* 2023;16:127–138
117. Gomez-Peralta F, Abreu C, Fernández-Rubio E, et al. Efficacy of a connected insulin pen cap in people with noncontrolled type 1 diabetes: a multicenter randomized clinical trial. *Diabetes Care* 2023;46:206–208
118. Cranston I, Jamdade V, Liao B, Newson RS. Clinical, economic, and patient-reported benefits of connected insulin pen systems: a systematic literature review. *Adv Ther* 2023;40:2015–2037
119. Danne TPA, Joubert M, Hartvig NV, Kaas A, Knudsen NN, Mader JK. Association between treatment adherence and continuous glucose monitoring outcomes in people with diabetes using smart insulin pens in a real-world setting. *Diabetes Care* 2024;47:995–1003
120. Bailey TS, Stone JY. A novel pen-based Bluetooth-enabled insulin delivery system with insulin dose tracking and advice. *Expert Opin Drug Deliv* 2017;14:697–703
121. Eiland L, McLarney M, Thangavelu T, Drincic A. App-based insulin calculators: current and future state. *Curr Diab Rep* 2018;18:123
122. Breton MD, Patek SD, Lv D, et al. Continuous glucose monitoring and insulin informed advisory system with automated titration and dosing of insulin reduces glucose variability in type 1 diabetes mellitus. *Diabetes Technol Ther* 2018;20:531–540
123. Bergenstal RM, Johnson M, Passi R, et al. Automated insulin dosing guidance to optimise insulin management in patients with type 2 diabetes: a multicentre, randomised controlled trial. *Lancet* 2019;393:1138–1148
124. Schneider JE, Parikh A, Stojanovic I. Impact of a novel insulin management service on non-insulin pharmaceutical expenses. *J Health Econ Outcomes Res* 2018;6:53–62
125. Huckvale K, Adomaviciute S, Prieto JT, Leow MK-S, Car J. Smartphone apps for calculating insulin dose: a systematic assessment. *BMC Med* 2015;13:106
126. Fan W, Deng C, Xu R, et al. Efficacy and safety of automated insulin delivery systems in patients with type 1 diabetes mellitus: a systematic review and meta-analysis. *Diabetes Metab J* 2025;49:235–251
127. Aleppo G, DeSalvo DJ, Lauand F, et al. Improvements in glycemic outcomes in 4738 children, adolescents, and adults with type 1 diabetes initiating a tubeless insulin management system. *Diabetes Ther* 2023;14:593–610
128. Gangiang G, Hua C, Hongyu S. Risk predictors of glycaemic control in children and adolescents with type 1 diabetes: a systematic review and meta-analysis. *J Clin Nurs* 2024;33:2412–2426
129. Howe CJ, Morone J, Hawkes CP, Lipman TH. Racial disparities in technology use in children with type 1 diabetes: a qualitative content analysis of parents' perspectives. *Sci Diabetes Self Manag Care* 2023;49:55–64
130. Tanenbaum ML, Commissariat PV, Wilmot EG, Lange K. Navigating the unique challenges of automated insulin delivery systems to facilitate effective uptake, onboarding, and continued use. *J Diabetes Sci Technol* 2025;19:47–53
131. Berghaeuser MA, Kapellen T, Heidtmann B, Haberland H, Klinkert C, Holl RW; German Working Group for Insulin Pump Treatment in Paediatric Patients. Continuous subcutaneous insulin infusion in toddlers starting at diagnosis of type 1 diabetes mellitus. A multicenter analysis of 104 patients from 63 centres in Germany and Austria. *Pediatr Diabetes* 2008;9:590–595
132. Peters AL, Ahmann AJ, Battelino T, et al. Diabetes technology-continuous subcutaneous insulin infusion therapy and continuous glucose monitoring in adults: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab* 2016;101:3922–3937
133. Karges B, Rosenbauer J, Stahl-Peche A, et al. Hybrid closed-loop insulin therapy and risk of severe hypoglycaemia and diabetic ketoacidosis in young people (aged 2–20 years) with type 1 diabetes: a population-based study. *Lancet Diabetes Endocrinol* 2025;13:88–96
134. Ucieklak D, Mrozinska S, Wojnarska A, Malecki MT, Klupa T, Matejko B. Insulin-induced lipohypertrophy in patients with type 1 diabetes mellitus treated with an insulin pump. *Int J Endocrinol* 2022;2022:9169296
135. Wong JC, Boyle C, DiMeglio LA, et al.; T1D Exchange Clinic Network. Evaluation of pump discontinuation and associated factors in the T1D Exchange Clinic Registry. *J Diabetes Sci Technol* 2017;11:224–232
136. Royston C, Hovorka R, Boughton CK. Closed-loop therapy: recent advancements and potential predictors of glycaemic outcomes. *Expert Opin Drug Deliv* 2025;22:875–892
137. Sherr JL, Hermann JM, Campbell F, et al.; T1D Exchange Clinic Network, the DPV Initiative, and the National Paediatric Diabetes Audit and the Royal College of Paediatrics and Child Health registries. Use of insulin pump therapy in children and adolescents with type 1 diabetes and its impact on metabolic control: comparison of results from three large, transatlantic paediatric registries. *Diabetologia* 2016;59:87–91
138. Commissariat PV, Boyle CT, Miller KM, et al. Insulin pump use in young children with type 1 diabetes: sociodemographic factors and parent-reported barriers. *Diabetes Technol Ther* 2017;19:363–369
139. Forlenza GP, Li Z, Buckingham BA, et al. Predictive low-glucose suspend reduces hypoglycemia in adults, adolescents, and children with type 1 diabetes in an at-home randomized crossover study: results of the PROLOG trial. *Diabetes Care* 2018;41:2155–2161
140. Wood MA, Shulman DI, Forlenza GP, et al. In-clinic evaluation of the MiniMed 670G System "suspend before low" feature in children with type 1 diabetes. *Diabetes Technol Ther* 2018;20:731–737
141. Beato-Vibora PI, Quirós-López C, Lázaro-Martín L, et al. Impact of sensor-augmented pump therapy with predictive low-glucose suspend function on glycemic control and patient satisfaction in adults and children with type 1 diabetes. *Diabetes Technol Ther* 2018;20:738–743
142. Brown SA, Beck RW, Raghinaru D, et al.; iDCL Trial Research Group. Glycemic outcomes of use of CLC versus PLGS in type 1 diabetes: a randomized controlled trial. *Diabetes Care* 2020;43:1822–1828
143. Akturk HK, Brown SA, Beck RW, et al. Registration and real-life studies on automated insulin delivery systems. *Diabetes Technol Ther* 2025;27:S48–S59
144. Gera S, Iyer J, Meighan S, March CA, Marks BE. A review of real-world evidence about the use of automated insulin delivery systems in people with type 1 diabetes. *Endocr Pract* 2025;31:1150–1161
145. Marrero DG, Parkin CG, Aleppo G, et al. The role of advanced technologies in improving diabetes outcomes. *Am J Manag Care* 2025;31:e102–e112
146. Kudva YC, Raghinaru D, Lum JW, et al.; 21QP Study Group. A randomized trial of automated insulin delivery in type 2 diabetes. *N Engl J Med* 2025;392:1801–1812
147. Pasquel FJ, Davis GM, Huffman DM, et al.; Omnipod 5 SECURE-T2D Consortium. Automated insulin delivery in adults with type 2 diabetes: a nonrandomized clinical trial. *JAMA Netw Open* 2025;8:e2459348
148. Boughton CK, Hovorka R. The role of automated insulin delivery technology in diabetes. *Diabetologia* 2024;67:2034–2044
149. Sherr JL, Cengiz E, Palerm CC, et al. Reduced hypoglycemia and increased time in target using closed-loop insulin delivery during nights with or without antecedent afternoon exercise in type 1 diabetes. *Diabetes Care* 2013;36:2909–2914
150. Berget C, Cobry E, Escobar E, et al. Sustained improvements in glycaemic and psychosocial outcomes for youth and caregivers using Omnipod 5 AID for 12 months. *Diabetes Obes Metab* 2025;27:4942–4949
151. Weissberg-Benchell J, Hessler D, Polonsky WH, Fisher L. Psychosocial impact of the bionic pancreas during summer camp. *J Diabetes Sci Technol* 2016;10:840–844
152. Troncone A, Bonfanti R, Iafusco D, et al. Evaluating the experience of children with type 1 diabetes and their parents taking part in an artificial pancreas clinical trial over multiple days in a diabetes camp setting. *Diabetes Care* 2016;39:2158–2164
153. Barnard KD, Wysocki T, Allen JM, et al. Closing the loop overnight at home setting: psychosocial impact for adolescents with type 1 diabetes and their parents. *BMJ Open Diabetes Res Care* 2014;2:e000025
154. Carlson AL, Sherr JL, Shulman DI, et al. Safety and glycemic outcomes during the MiniMed advanced hybrid closed-loop system pivotal trial in adolescents and adults with type 1 diabetes. *Diabetes Technol Ther* 2022;24:178–189
155. Weissberg-Benchell J, Vesco AT, Shapiro J, et al. Psychosocial impact of the insulin-only iLet bionic pancreas for adults, youth, and caregivers of youth with type 1 diabetes. *Diabetes Technol Ther* 2023;25:705–717
156. Vallivaara H-L, Leppänen HA, Mustonen J, Saari A, Martikainen J, Rintamäki RM. Long-term

- health economic evaluation of automated insulin delivery system compared with continuous subcutaneous insulin infusion pumps and CGM in a real-world setting in Finnish paediatric and adult individuals with type 1 diabetes. *Diabetes Obes Metab* 2025;27:4793–4801
157. Considine EG, Sherr JL. Real-world evidence of automated insulin delivery system use. *Diabetes Technol Ther* 2024;26:53–65
158. Forlenza GP, DeSalvo DJ, Aleppo G, et al. Real-world evidence of Omnipod 5 automated insulin delivery system use in 69,902 people with type 1 diabetes. *Diabetes Technol Ther* 2024;26:514–525
159. Amigó J, Ortiz-Zúñiga Á, de Urbina AMO, et al. Switching from treatment with sensor augmented pump to hybrid closed loop system in type 1 diabetes: impact on glycemic control and neuropsychological tests in the real world. *Diabetes Res Clin Pract* 2023;201:110730
160. Chico A, Navas de Solís S, Lainez M, Rius F, Cuesta M. Efficacy, safety, and satisfaction with the Accu-Chek Insight with Diabeloop closed-loop system in subjects with type 1 diabetes: a multicenter real-world study. *Diabetes Technol Ther* 2023;25:242–249
161. Benhamou P-Y, Adenis A, Lebbad H, et al. One-year real-world performance of the DBLG1 closed-loop system: data from 3706 adult users with type 1 diabetes in Germany. *Diabetes Obes Metab* 2023;25:1607–1613
162. Benhamou P-Y, Adenis A, Lablanche S, et al. First generation of a modular interoperable closed-loop system for automated insulin delivery in patients with type 1 diabetes: lessons from trials and real-life data. *J Diabetes Sci Technol* 2023;17:1433–1439
163. Beck RW, Kanapka LG, Breton MD, et al. A meta-analysis of randomized trial outcomes for the t:slim X2 insulin pump with Control-IQ technology in youth and adults from age 2 to 72. *Diabetes Technol Ther* 2023;25:329–342
164. Grassi B, Gómez AM, Calliari LE, et al. Real-world performance of the MiniMed 780G advanced hybrid closed loop system in Latin America: substantial improvement in glycaemic control with each technology iteration of the MiniMed automated insulin delivery system. *Diabetes Obes Metab* 2023;25:1688–1697
165. Forlenza GP, Carlson AL, Galindo RJ, et al. Real-world evidence supporting tandem Control-IQ hybrid closed-loop success in the Medicare and Medicaid type 1 and type 2 diabetes populations. *Diabetes Technol Ther* 2022;24:814–823
166. Grunberger G, Rosenfeld CR, Bode BW, et al. Effectiveness of V-Go for patients with type 2 diabetes in a real-world setting: a prospective observational study. *Drugs Real World Outcomes* 2020;7:31–40
167. Raval AD, Nguyen MH, Zhou S, Grabner M, Barron J, Quimbo R. Effect of V-Go versus multiple daily injections on glycemic control, insulin use, and diabetes medication costs among individuals with type 2 diabetes mellitus. *J Manag Care Spec Pharm* 2019;25:1111–1123
168. Winter A, Lintner M, Knezevich E. V-Go insulin delivery system versus multiple daily insulin injections for patients with uncontrolled type 2 diabetes mellitus. *J Diabetes Sci Technol* 2015;9:1111–1116
169. Bergenstal RM, Peyrot M, Dreon DM, et al.; Calibra Study Group. Implementation of basal-bolus therapy in type 2 diabetes: a randomized controlled trial comparing bolus insulin delivery using an insulin patch with an insulin pen. *Diabetes Technol Ther* 2019;21:273–285
170. Braune K, Lal RA, Petruželková L, et al.; OPEN International Healthcare Professional Network and OPEN Legal Advisory Group. Open-source automated insulin delivery: international consensus statement and practical guidance for health-care professionals. *Lancet Diabetes Endocrinol* 2022;10:58–74
171. Lum JW, Bailey RJ, Barnes-Lomen V, et al. A real-world prospective study of the safety and effectiveness of the loop open source automated insulin delivery system. *Diabetes Technol Ther* 2021;23:367–375
172. Braune K, Gajewska KA, Thieffry A, et al. Why #WeAreNotWaiting-motivations and self-reported outcomes among users of open-source automated insulin delivery systems: multinational survey. *J Med Internet Res* 2021;23:e25409
173. Burnside MJ, Lewis DM, Crockett HR, et al. Open-source automated insulin delivery in type 1 diabetes. *N Engl J Med* 2022;387:869–881
174. Petruželkova L, Neuman V, Plachy L, et al. First use of open-source automated insulin delivery AndroidAPS in full closed-loop scenario: Pancreas4ALL randomized pilot study. *Diabetes Technol Ther* 2023;25:315–323
175. Braune K, Hussain S, Lal R. The first regulatory clearance of an open-source automated insulin delivery algorithm. *J Diabetes Sci Technol* 2023;17:1139–1141
176. Phillip M, Bergenstal RM, Close KL, et al. The digital/virtual diabetes clinic: the future is now—recommendations from an international panel on diabetes digital technologies introduction. *Diabetes Technol Ther* 2021;23:146–154
177. Fleming GA, Petrie JR, Bergenstal RM, Holl RW, Peters AL, Heinemann L. Diabetes digital app technology: benefits, challenges, and recommendations. A consensus report by the European Association for the Study of Diabetes (EASD) and the American Diabetes Association (ADA) Diabetes Technology Working Group. *Diabetes Care* 2020;43:250–260
178. Wong JC, Izadi Z, Schroeder S, et al. A pilot study of use of a software platform for the collection, integration, and visualization of diabetes device data by health care providers in a multidisciplinary pediatric setting. *Diabetes Technol Ther* 2018;20:806–816
179. Chao DY, Lin TM, Ma W-Y. Enhanced self-efficacy and behavioral changes among patients with diabetes: cloud-based mobile health platform and mobile app service. *JMIR Diabetes* 2019;4:e11017
180. Sepah SC, Jiang L, Peters AL. Translating the diabetes prevention program into an online social network: validation against CDC standards. *Diabetes Educ* 2014;40:435–443
181. Kaufman N, Ferrin C, Sugrue D. Using digital health technology to prevent and treat diabetes. *Diabetes Technol Ther* 2019;21:S79–S94
182. Öberg U, Isaksson U, Jutterström L, Orre CJ, Hörnsten Å. Perceptions of persons with type 2 diabetes treated in Swedish primary health care: qualitative study on using eHealth services for self-management support. *JMIR Diabetes* 2018;3:e7
183. Bollyky JB, Bravata D, Yang J, Williamson M, Schneider J. Remote lifestyle coaching plus a connected glucose meter with certified diabetes educator support improves glucose and weight loss for people with type 2 diabetes. *J Diabetes Res* 2018;2018:3961730
184. Wilhide Ili CC, Peeples MM, Anthony Kouyaté RC. Evidence-based mHealth chronic disease mobile app intervention design: development of a framework. *JMIR Res Protoc* 2016;5:e25
185. Dixon RF, Zisser H, Layne JE, et al. A virtual type 2 diabetes clinic using continuous glucose monitoring and endocrinology visits. *J Diabetes Sci Technol* 2020;14:908–911
186. Yang Y, Lee EY, Kim H-S, Lee S-H, Yoon K-H, Cho J-H. Effect of a mobile phone-based glucose-monitoring and feedback system for type 2 diabetes management in multiple primary care clinic settings: cluster randomized controlled trial. *JMIR Mhealth Uhealth* 2020;8:e16266
187. Levine BJ, Close KL, Gabbay RA. Reviewing U.S. connected diabetes care: the newest member of the team. *Diabetes Technol Ther* 2020;22:1–9
188. McGill DE, Volkening LK, Butler DA, Wasserman RM, Anderson BJ, Laffel LM. Text-message responsiveness to blood glucose monitoring reminders is associated with HbA1c benefit in teenagers with type 1 diabetes. *Diabet Med* 2019;36:600–605
189. Shen Y, Wang F, Zhang X, et al. Effectiveness of internet-based interventions on glycemic control in patients with type 2 diabetes: meta-analysis of randomized controlled trials. *J Med Internet Res* 2018;20:e172
190. Kumbara AB, Iyer AK, Green CR, et al. Impact of a combined continuous glucose monitoring-digital health solution on glucose metrics and self-management behavior for adults with type 2 diabetes: real-world, observational study. *JMIR Diabetes* 2023;8:e47638
191. U.S. Food and Drug Administration. Self-Monitoring Blood Glucose Test Systems for Over-the-Counter Use. Guidance for Industry and Food and Drug Administration Staff, September 2020. Accessed 19 August 2025. Available from <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/self-monitoring-blood-glucose-test-systems-over-counter-use>
192. U.S. Food and Drug Administration. Blood Glucose Monitoring Test Systems for Prescription Point-of-Care Use: Guidance for Industry and Food and Drug Administration Staff, September 2020. Accessed 15 August 2025. Available from <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/blood-glucose-monitoring-test-systems-prescription-point-care-use>
193. Pardo S, Simmons DA. The quantitative relationship between ISO 15197 accuracy criteria and mean absolute relative difference (MARD) in the evaluation of analytical performance of self-monitoring of blood glucose (SMBG) systems. *J Diabetes Sci Technol* 2016;10:1182–1187